

Comparison of Eye-Tracking Representations for Engagement Inference in Office Workers

[To COMPLETE: Author name]

[To COMPLETE: Institution]

[To COMPLETE: City], [To COMPLETE: Country]

author@example.org

Abstract

Keywords

eye tracking, engagement inference, representation learning, GazeMAE, MOMENT, subject generalisation

1 Introduction

- Engagement inference in office work
- Eye tracking: gaze behaviour, pupil dynamics, behavioural evidence
- Practical representation choice for eye-tracking windows
- Raw temporal signals: information retention, high dimensionality
- Handcrafted features: domain knowledge, interpretability
- Learned embeddings: transfer promise, preprocessing and deployment cost
- Gaze-specific GazeMAE; general time-series MOMENT
- Evidence gap: like-for-like comparison on identical windows and labels
- Unseen-person generalisation; subject-disjoint evaluation
- Research question: representation reliability for self-reported engagement in unseen office workers
- Controlled comparison: raw, handcrafted, GazeMAE, MOMENT
- Frozen pretrained encoders; trained prediction head only
- Contribution: leakage-safe LOSO evaluation
- Contribution: practical representation-selection guidance

2 Related Work

Eye tracking has long been used as behavioural evidence for internal user states. Pupil dilation is a classical correlate of processing load [5], and fixation, saccade, blink and pupil statistics computed over short windows remain the dominant input to models of workload, affect and engagement. Our own prior work illustrates both the appeal and the limits of this pipeline: handcrafted ocular features supported coarse affective distinctions [3], yet a systematic evaluation across several prediction targets showed that task type is recognised far more reliably than graded workload, and that reported accuracy depends strongly on how the prediction target and the validation split are defined [2]. Feature engineering encodes useful domain knowledge, but the resulting performance is bounded by what a fixed feature set captures.

Learned representations offer an alternative. GazeMAE pre-trains temporal autoencoders on raw position and velocity signals and shows that the resulting gaze-specific embeddings support downstream classification through a simple linear head [1]. In

parallel, general-purpose time-series foundation models such as MOMENT provide a frozen multichannel encoder intended to transfer across domains without task-specific pretraining [4]. Both are presented as replacements for hand-designed features, and both are typically evaluated on curated benchmarks rather than on modest, subject-dependent recordings of spontaneous behaviour. How much of that benefit survives when such encoders are applied out of the box in this setting remains open.

Evaluation design is a further concern. Windows drawn from the same person are highly dependent, so random window splits and preprocessing fitted on the full dataset inflate reported performance, an effect documented in detail for other biosignal domains [6]. Comparisons across representations are interpretable only when every candidate is evaluated on identical windows and labels under a subject-disjoint protocol with fold-local preprocessing.

We therefore extend our earlier feature-based analyses to learned representations. This preliminary study is an empirical comparison rather than a new encoder or fine-tuning method: raw signals, handcrafted features, GazeMAE and MOMENT embeddings are evaluated on the same windows, the same engagement target and the same leave-one-subject-out folds.

3 Methodology

3.1 Data and Eye-Tracking Representations

- TrustMe study; office-worker recordings
- Participant count, sessions, recording duration, sampling rate, exclusions
- Tobii channels: left/right pupil size, gaze x/y , average distance
- q5 engagement item: “How immersed in your work did you feel just now?”
- q5 response scale and observed raw-label distribution
- Window duration, overlap, temporal ordering
- Common window_uid intersection across all four representations
- Labelled windows only; final subject and window counts
- Validity fields; missing-value and quality-control policy
- Display-resolution metadata; per-recording x/y normalization to $[0, 1]$
- Out-of-screen coordinates: retained for modelling policy; excluded from descriptive heatmap
- Representation order: raw, handcrafted features, GazeMAE embeddings, MOMENT embeddings
- Raw: five channels, 120 samples per channel, truncate/pad policy, final dimensionality
- Handcrafted: feature families, metadata removal, missingness filtering, correlation filtering
- GazeMAE: checkpoint, inputs, embedding selection, dimensionality, frozen encoder
- MOMENT: checkpoint, inputs, pooling, dimensionality, frozen encoder

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- **Figure 1 — normalized gaze-density map:** screen-coordinate coverage; recording-quality context; non-predictive descriptive evidence

3.2 Experimental Design

- Primary task: binary q5 engagement inference
- LOSO target construction: training-subject mean centring; held-out subject global-training-mean centring
- Diagnostic visualization label only: raw q5; binary q5 above global processed-data median; no use in model training or metrics
- Same aligned cohort, target definition, and folds across representations
- Majority-class baseline
- Random Forest: 300 trees, balanced class weights, fixed seed
- MLP: GELU, layer normalization, Adam, early stopping, fixed seed
- Representation-by-model grid; robustness check rather than model-comparison claim
- Fold-local handcrafted-feature selection and imputation
- Fold-local median imputation; train-fit standardization; held-out-subject transform
- No test-subject information in preprocessing parameters
- Accuracy: primary metric
- Accuracy delta and balanced-accuracy delta: paired majority-baseline comparison within LOSO folds
- Macro-F1 and ROC-AUC: secondary discrimination metrics
- Subject-macro aggregation
- One-class test folds: retained accuracy; undefined balanced accuracy, macro-F1, and AUC; reported valid-fold counts
- Persistence baseline: contextual within-subject temporal check; text only; not comparable with unseen-subject main table

4 Results

- **Table 1 — primary LOSO results:** majority baseline; raw, handcrafted, GazeMAE, MOMENT; Random Forest and MLP
- Table columns: representation; model; accuracy (Δ majority); balanced accuracy (Δ majority); macro-F1; ROC-AUC
- Table order: majority baseline; raw; handcrafted; GazeMAE; MOMENT
- Subject-macro estimates; valid-fold counts in caption or table note
- Bold: best value per metric; no overemphasis on small model differences
- Main result: highest accuracy; absolute and majority-baseline improvement
- Representation result: raw and handcrafted performance relative to frozen GazeMAE and MOMENT embeddings
- Model result: consistency of representation ranking across Random Forest and MLP
- One-class held-out-subject treatment; metric availability
- Persistence-check result: contextual temporal predictability; separate protocol; no direct ranking against LOSO models
- **Figure 2 — selected projection:** one representation-method combination; raw q5 continuous colour scale

- Figure-selection criterion: clearest engagement gradient or clearest absence of label separation
- Figure interpretation: exploratory geometry; mixed colours as limited label separability; no causal or performance claim

5 Conclusion

- Discussion part:
- Direct answer to the research question
- Frozen learned embeddings: out-of-the-box utility relative to raw and handcrafted alternatives
- Representation effect; secondary role of classifier choice
- Practical recommendation: accuracy, preprocessing burden, storage, computation, interpretability
- Implication of non-improving embeddings: task-pretraining mismatch; need for domain adaptation or fine-tuning
- Projection and heatmap: descriptive support only; no separability proof
- Persistence result: temporal autocorrelation; distinct from cross-person generalisation
- Limitations: single dataset, office-worker sample, self-report noise, binary target construction
- Limitations: frozen checkpoints, fixed windowing, quality-control choices, limited subject count
- Future work: multi-dataset validation, calibrated personalisation, adaptation/fine-tuning, alternative target formulations
- Conclusion part:
- Controlled four-representation comparison for unseen-worker engagement inference
- Final best-performing representation and headline accuracy
- Frozen-encoder conclusion; out-of-the-box deployment recommendation
- Concise practitioner guidance: representation choice under limited labelled data

Acknowledgements

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